Saturday, January 31, 2026 4:46 PM

$V_1 = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$ find V_2 orthogonal to V_1

$$V_1 \cdot V_2 = 0$$

$$\boxed{\vec{V_1} \cdot \vec{V_2} = |V_1||V_2|\cos(\theta)}$$

$\theta = 90$

$\cos(90) = 0$

$V_2 = \begin{bmatrix} a \\ b \end{bmatrix}$

$$\begin{bmatrix} \frac{3}{5} & \frac{4}{5} \end{bmatrix} \cdot \begin{bmatrix} a & b \end{bmatrix} = 0$$

1×2 $\quad$ 2×1

$$\frac{3}{5}(a) + \frac{4}{5}b = 0$$

$$\boxed{a = -\frac{4}{5}b\left(\frac{5}{3}\right)} \qquad \boxed{a = -\frac{4}{3}b}$$

$b = 1$

$a = -\frac{4}{3}$

$$V_2 = \begin{bmatrix} -\frac{4}{3} \\ 1 \end{bmatrix} \not\propto \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

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Assume $V_2 = \begin{bmatrix} -4/5 \end{bmatrix}$

Assume $v_2 = \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}$

$= -\frac{12}{25} + \frac{12}{25}$

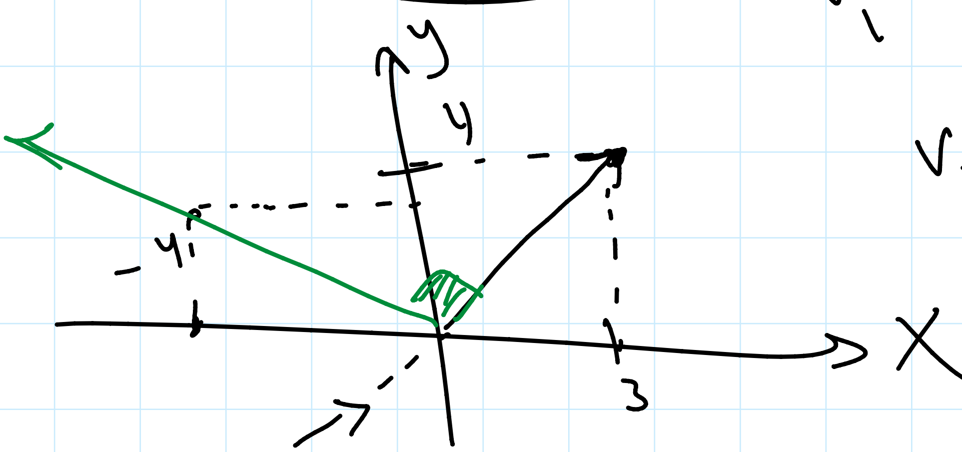
$= 0$ to $= 0$

$V_1 = \begin{bmatrix} a \\ b \end{bmatrix} \Rightarrow V_2 = \begin{bmatrix} -b \\ a \end{bmatrix} = \begin{bmatrix} b \\ -a \end{bmatrix}$

$\theta = 90$

$v_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$v_2 = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$



$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}$

A^TA & AA^T

$$A^T \cdot A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & d \end{bmatrix} \begin{array}{c} 4 \times 3 \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & d \end{bmatrix} \end{array}$$

$$\begin{bmatrix} 0 & 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$\lambda_3 = 1, \lambda_1 = 4, \quad \lambda_2 = 4$

$\lambda_i = 4, 4, 1$

$\sigma_i = 2, 2, 2$

$\sigma_1 = 2$

$\sigma_2 = 2$

$\sigma_3 = 1$

$$\Sigma = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}_{4 \times 3}$$

$V \Rightarrow$ matrix $\quad |A^TA - \lambda I| = 0$

$$|A^TA - = \begin{bmatrix} 4-\lambda & 0 & 0 \\ 0 & 4-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{bmatrix}$$

at $\lambda = 4$

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -3 & 0 \end{array}\right]$$

$-3z = 0$

$-3z = 0$

$z = 0$

$x = r$

$y = s$

$r = 1$

$s = 0$

$$V_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} r + \cancel{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} s}$$

$$V_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

at $\lambda_2 = 4$

$$V_2 = \cancel{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} r} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} s$$

$r = 0$

$$V_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

at $\lambda = 1$

$$\left[\begin{array}{ccc|c} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$3x = 0$

$x = 0$

$y = 0$

$z = r$

$$V_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r \quad r = 1$$

$$V_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$V_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad V_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad V_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$V = [e_1 \vdots e_2 \vdots e_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

<u>then</u> you have to Calculate U_i

$$U_i = \frac{1}{\sigma_i} A V_i$$

$$U_1 = \frac{1}{2} \underset{4 \times 3}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}} \underset{3 \times 1}{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}} = \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$U_1 = \begin{bmatrix} 0.5 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

simlarly $U_2 = \frac{1}{2} A \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ .05 \\ 0 \\ 0 \end{bmatrix}$

$$U_3 = \frac{1}{1}\begin{bmatrix}A\end{bmatrix}\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

U_1 ✓ U_2 ✓ U_3 ✓ U_4???

$$y = \frac{3x+1}{2x+6} = \frac{3\left[x+\frac{1}{3}\right]}{2x+6}$$

$$= \frac{\frac{3}{2}\left[2x+\frac{2}{3}\right]}{2x+6}$$

$$= \frac{\frac{3}{2}\left[(2x+6) - 6 + \frac{2}{3}\right]}{2x+6}$$

$$\frac{A+B}{C} = \frac{A}{C} + \frac{B}{C}$$

$$= \frac{3}{2}\left[\frac{2x+6-\frac{16}{3}}{2x+6}\right]$$

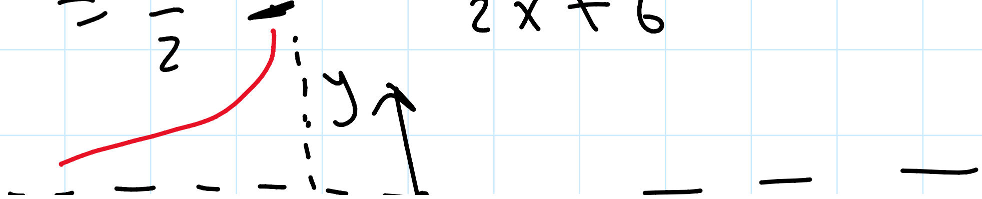
$$= \frac{3}{2}\left[1 - \frac{\frac{16}{3}}{2x+6}\right]$$

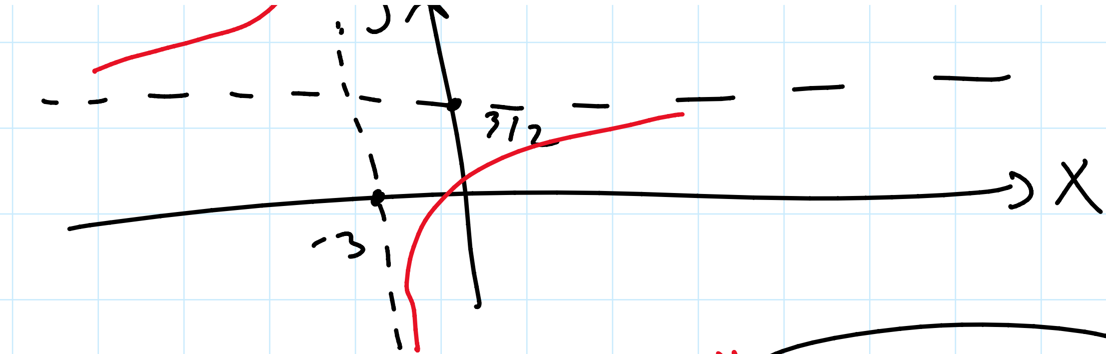
$$= \frac{3}{2} - \frac{8}{2x+6}$$

$2x+6=0$

$2x=-6$

$\boxed{x=-3}$





$x \leftrightarrow y$

$$y = \frac{3x+1}{2x+6} \Rightarrow x = \frac{3y+1}{2y+6}$$

$y = ??$

$$3y+1 = 2xy + 6x$$

$$3y - 2xy = 6x - 1$$

$$y[3-2x] = 6x - 1$$

$$y = \frac{6x-1}{3-2x}$$

$$f^{-1}(x) = \frac{6x-1}{3-2x}$$

$$f \circ f^{-1}(x) = x$$

$$f^{-1} \circ f(x) = x$$

$f \circ f(x) = x$